

Ritabrata Ghosh

📍 Bangalore, India ✉ ritabrataghosh09@gmail.com 📞 +91 7365956148 📁 Portfolio in LinkedIn 🐙 Github

Skills

Languages: C/C++, JavaScript, TypeScript, Python

Frontend: React.js, Next.js

Backend: Node.js, Express.js, trpc

Databases & Infrastructure: PostgreSQL, MongoDB, Prisma, Redis, Kafka, Websockets, Docker, AWS, CI/CD

AI & Developer Tools: OpenAI API, Gemini API, RAG Pipelines, PineconeDB, LangChain, LangGraph

Education

Visvesvaraya Technological University

B.E. in Computer Science and Engineering

Aug 2023 - June 2027

CGPA:8.5/10

Experience

Open Source Contributor, JSON Schema Organization

Nov 2025 - Present

- Contributed to **json-schema-org**, **Sourcemeta**, and **Hyperjump** by improving JSON Schema validation workflows, CLI tooling, and developer experience.
- Enhanced **error categorization**, **lint ordering by source location**, and **exit code classification** to ensure standards-compliant schema validation behavior.
- Implemented features such as **pagination**, **state persistence**, search state management across actively maintained production-scale repositories.

Projects

CollabDraw: Real-Time Collaborative Whiteboard



(Next.js, Node.js, WebSockets, Kafka, PostgreSQL, Turborepo, Rough.js)

- Developed a real-time collaborative whiteboard using **WebSockets** with scalable room-based architecture and synchronized state management.
- Integrated **Kafka** for event-driven, low-latency chat messaging within sessions for scalability.
- Implemented drawing engine with **Rough.js** and structured a scalable monorepo using **Turborepo**.

BrainDoc: AI-Powered Document Intelligence Platform



(Next.js, Node.js, AWS S3, PostgreSQL, PineconeDB, RAG Pipeline, OpenAI API,

YouTube Data API)

- Built an AI-driven document query system using a **Retrieval-Augmented Generation (RAG)** pipeline for context-aware responses.
- Leveraged **Pinecone vector database** for semantic search and scalable embedding retrieval.
- Designed full-stack architecture with persistent chat history and contextual video retrieval integration and document upload on **AWS S3**.

Planova: AI-Powered Travel Planning System



(React, Node.js, Express, MongoDB, Gemini API, Google Places API, REST APIs)

- Developed an AI-driven itinerary generation system using **Gemini API** for personalized, constraint-aware travel planning.
- Integrated **Google Places API** for real-time location insights and hotel recommendations.

Achievements

- 1st Place: SJBIT Hackathon AXIOM 2025** - Won 50,000 prize for developing an innovative hardware based healthcare solution.
- 1st Place: Dr. TTIT Nexathon 2025** - Secured 20,000 prize for building a high-impact technical prototype under time constraints.
- 2nd Place: Global Ideathon 2025** - Awarded 15,000 prize for presenting a scalable and practical innovation concept.